

2a. The advantages to the Array representation are that it has compact memory usage, and is able to efficiently access items through an index, but its downsides are that it may waste memory when we use zero coefficients and can be inefficient when adding or removing items due to resizing. The advantages of List representation is that it efficiently stores non-zero coefficients and is flexible adding or removing terms without any need to resize, and its disadvantages are that it uses slightly more memory due to references of objects and the possibility of a slower ability to access coefficients by exponent.

2b. The `checkRep()` function is called at the end of the constructors because it ensures that the constructors follow the rules of the invariants at the end of their process. It is not called anywhere else because `RatPoly` was implemented in such a way that instances of the class preserve the integrity of their invariants, which prevents the need for the usage of the `checkRep()` method.